Question 6

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.1		if no answer in answer space allow answer in **Table 6**		AO2 4.1.3.1
	$\frac{9.96 \times 10^{-3}}{1.35 \times 10^{-4}}$	allow $\frac{0.00996}{0.000135}$	1	
	73.77…		1	
	74 (:1)	allow a correctly derived whole number from an incorrect calculation do **not** accept if unit is given	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.2	as size increases, (surface area to volume) ratio decreases	allow they are inversely proportional **or** they are negatively correlated allow as one increases, the other decreases allow as size decreases, (surface area to volume) ratio increases	1	AO2 4.1.3.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.3		allow converse for **B** throughout		
	D has a smaller surface area to volume ratio (than **B**)		1	AO3
	(so) diffusion distance is too large (to meet demands of cells / organism)	allow (so) diffusion takes too long (to meet demands of cells / organism)	1	AO2 4.1.3.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.4		allow converse for **E** throughout		AO3
	D has a larger surface area to volume ratio **and** so will lose heat more quickly (per unit volume than **E**)	allow **D** has a larger surface area to volume ratio **and** so temperature of **D** will drop more quickly ignore **E** loses more heat (overall)	1	AO2
	(**D**) requires greater rate of respiration		1	AO2
	(as) respiration is a (large) part of metabolism		1	AO2
	(so) need to generate more <u>heat</u> (to keep itself warm)	allow (so) needs to release more <u>heat</u> (to keep itself warm) do **not** accept energy produced / made / created	1	4.1.3.1 4.4.2.1 4.4.2.3

Question	Answers	Mark	AO / Spec. Ref.
06.5	**Level 2:** Scientifically relevant facts, events or processes are identified and given in detail to form an accurate account.	3−4	AO1 4.1.3.1 4.1.3.3 4.2.2.2
	Level 1: Facts, events or processes are identified and simply stated but their relevance is not clear.	1−2	
	No relevant content	0	

	Indicative content
	
	• **both** have a large surface area
	  ○ to maximise diffusion
	
	• **both** have thin walls **or** have walls that are one cell thick
	  ○ to reduce diffusion distance / time
	
	• **both** are in close proximity to blood supply
	  ○ to reduce diffusion distance / time
	
	• **both** have a good blood supply **or** both have a capillary network
	  ○ to maintain concentration gradient
	
	• villi have microvilli
	  ○ to (further) increase surface area
	
	• cells of villi contain many mitochondria
	  ○ for active transport
	
	For **Level 2** reference to functions of structural details of **both** alveoli and villi is required.		

Total Question 6		**14**